Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

AWS Certified Machine Learning Engineer Associate Training

TGS Ref No: TGS-2024049340

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 14

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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
12	23 August 2026	Superseded release — 2-day, 147-slide deck; eight labs mapped to the four MLA-C01 exam domains.	Tertiary Infotech Academy Pte Ltd
13	23 August 2026	Superseded release — concept-led 2-day rebuild (105 slides) with the eight labs; per-lab click-by-click steps moved from the deck into the Learner Guide and labs.	Tertiary Infotech Academy Pte Ltd
14	23 August 2026	Three-day rebuild to the verified authoritative duration (3 days / 24 hours = 22 learning hours + Written Assessment 1h + Case Study 1h, reconciling the LMS-TMS trainingHours=22). Expanded to twelve self-contained hands-on labs — exactly four unique labs per day (Day 1 Labs 1–4, Day 2 Labs 5–8, Day 3 Labs 9–12) — mapped to the four MLA-C01 exam domains. Each day totals 480 scheduled minutes (excluding lunch, including tea); the Slides column cites the exact current deck footer ranges for assessment traceability.	Tertiary Infotech Academy Pte Ltd

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Course Information

Course Title	AWS Certified Machine Learning Engineer Associate Training
WSQ Course Reference	TGS-2024049340
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Duration	3 days · 24 course hours (22 learning hours + 2 assessment hours)
Duration reconciliation	The registration page lists 3 days / 24 hours; the LMS-TMS trainingHours=22. These reconcile as 22 learning hours + Written Assessment 1h + Case Study 1h = 24 scheduled course hours.
Daily Timing	9:30 am – 6:30 pm (1-hour lunch; two 15-min tea breaks counted within the 8 scheduled hours per day)
Mode	Instructor-led classroom training with hands-on AWS labs (Console, CloudShell/CLI, Amazon SageMaker AI)
Hands-on Labs	Twelve self-contained labs — exactly four unique labs per day
Certification Alignment	AWS Certified Machine Learning Engineer - Associate (MLA-C01)
Trainer	Dr Alfred Ang

Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Prepare, ingest (batch and streaming), transform, engineer features for, label and govern data for machine learning on AWS using Amazon S3, AWS Glue, Athena, Kinesis, SageMaker Data Wrangler, Feature Store, Ground Truth and Clarify.
- LO2: Select a modelling approach and train, debug, tune, evaluate and register models with Amazon SageMaker AI, SageMaker JumpStart, Amazon Bedrock and AWS AI services.
- LO3: Deploy real-time, serverless, asynchronous and batch inference and orchestrate MLOps pipelines with SageMaker Pipelines, Step Functions, CodePipeline, EventBridge and Infrastructure as Code.
- LO4: Monitor, secure and cost-optimise ML solutions with SageMaker Model Monitor, CloudWatch, IAM, KMS, VPC and AWS cost tooling, and prepare for the AWS MLA-C01 certification exam.

Assessment

- Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book. Knowledge questions trace to specific course slides.

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- Case Study (CS) — an applied AWS ML scenario, 1 hour, open book. Learners analyse a business scenario and design and justify an end-to-end ML solution mapped to the four MLA-C01 domains and the twelve hands-on labs.
- Format: Open Book — course slides, Learner Guide and approved materials only.
- The final assessment is conducted on Day 3 (Written Assessment then Case Study) and totals 2 assessment hours.
- A minimum of 75% attendance is required to be eligible for assessment and funding.

Note: the Slides column in the schedule below references the exact slide range in the current course deck, so every knowledge question can be traced to specific slides and every case-study task to the relevant labs.

Course Schedule

Day 1 — Data Preparation for Machine Learning

Time	Duration	Topic / Activity	Slides
9:30–10:00	30 min	Welcome, course introduction, ground rules, mandatory AM digital attendance (TRAQOM), course material download, learning outcomes and assessment flow	1–13
10:00–10:40	40 min	**AWS machine learning engineering foundations: the ML lifecycle, service stack, data types/formats and the four MLA-C01 exam domains**	14–20
10:40–10:55	15 min	Tea break (within instructional time)	-
10:55–11:30	35 min	**Domain 1 key concepts: data formats, S3 ML data lake, encryption, storage classes, IAM access and batch/streaming ingestion**	21–37
11:30–12:30	60 min	Lab 1: Account Safety and IAM Baseline	38–41
12:30–13:30	60 min	Lab 2: S3 ML Data Lake and Data Formats	42–45
13:30–14:30	60 min	Lunch break	-
14:30–15:55	85 min	Lab 3: Batch and Streaming Ingestion with Glue, Athena and Kinesis	46–49

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15:55–16:10	15 min	Tea break (within instructional time)	-
16:10–17:50	100 min	Lab 4: Feature Engineering, Feature Store, Labelling and Data Quality	50–53
17:50–18:15	25 min	**Domain 1 consolidation: feature engineering, Feature Store, labelling, data quality and exam-decision patterns**	21–37
18:15–18:30	15 min	Day 1 recap, Q&A and PM digital attendance (TRAQOM)	54

Total scheduled time: 480 minutes (8 hours), inclusive of the two 15-minute tea breaks and exclusive of the 1-hour lunch.

Day 2 — ML Model Development

Time	Duration	Topic / Activity	Slides
9:30–9:50	20 min	Day 1 recap and mandatory AM digital attendance (TRAQOM)	54
9:50–10:25	35 min	**Domain 2 key concepts: modelling approach, algorithms, training & debugging, tuning, experiments, Model Registry and evaluation**	55–72
10:25–10:40	15 min	Tea break (within instructional time)	-
10:40–12:00	80 min	Lab 5: Modelling Approach, Algorithms and Managed AI Services	73–76
12:00–13:00	60 min	Lab 6: SageMaker Training Jobs and Debugger	77–80
13:00–14:00	60 min	Lunch break	-
14:00–15:30	90 min	Lab 7: Automatic Model Tuning, Experiments and Model Registry	81–84
15:30–15:45	15 min	Tea break (within instructional time)	-
15:45–17:15	90 min	Lab 8: Model Evaluation, Metrics, Bias and Explainability	85–88
17:15–18:15	60 min	**Domain 2 consolidation: metrics, bias & explainability and modelling exam-**	55–72

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		decision patterns	
18:15–18:30	15 min	Day 2 recap, Q&A and PM digital attendance (TRAQOM)	89

Total scheduled time: 480 minutes (8 hours), inclusive of the two 15-minute tea breaks and exclusive of the 1-hour lunch.

Day 3 — Deployment, Monitoring, Security & Exam Review

Time	Duration	Topic / Activity	Slides
9:30–9:50	20 min	Day 2 recap and mandatory AM digital attendance (TRAQOM)	89
9:50–10:25	35 min	**Domain 3 key concepts: inference options, endpoints & auto scaling, MLOps pipelines, CI/CD and Infrastructure as Code**	90–105
10:25–10:40	15 min	Tea break (within instructional time)	-
10:40–12:00	80 min	Lab 9: Deployment Options, Endpoints, Batch Transform and Auto Scaling	106–109
12:00–13:00	60 min	Lab 10: MLOps Pipelines, CI/CD, Orchestration and IaC	110–113
13:00–14:00	60 min	Lunch break	-
14:00–14:25	25 min	**Domain 4 key concepts: model monitoring & drift, observability, security, governance and cost optimisation**	115–129
14:25–15:20	55 min	Lab 11: Model Monitoring, Drift and Observability	130–133
15:20–15:35	15 min	Tea break (within instructional time)	-
15:35–16:15	40 min	Lab 12: Security, Governance, Cost Optimisation and Exam Review	134–137
16:15–16:30	15 min	**Briefing for Assessment (open-book rules, integrity, submission and appeal)**	11
16:30–17:30	60 min	**Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book (knowledge, traces to course**	12

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		slides)	
17:30–18:30	60 min	**Case Study (CS) — applied AWS ML scenario, 1 hour, open book (traces to the twelve labs); Assessment & PM digital attendance**	146–155

Total scheduled time: 480 minutes (8 hours), inclusive of the two 15-minute tea breaks and exclusive of the 1-hour lunch — 360 min learning + 120 min assessment (WA 60 min + CS 60 min).

Course total: 22 learning hours + 2 assessment hours (Written Assessment 1h + Case Study 1h) = 24 course hours across 3 days.

Lab Reference (aligned to MLA-C01 exam domains)

Exam domain	Weighting	Labs	Slides
Domain 01: Data Preparation for Machine Learning	28%	Lab 1, Lab 2, Lab 3, Lab 4	38–53
Domain 02: ML Model Development	26%	Lab 5, Lab 6, Lab 7, Lab 8	73–88
Domain 03: Deployment and Orchestration of ML Workflows	22%	Lab 9, Lab 10	106–113
Domain 04: ML Solution Monitoring, Maintenance and Security	24%	Lab 11, Lab 12	130–137

Daily Lab Plan (four unique labs per day)

Day	Theme	Labs (exactly four)
Day 1	Data Preparation for Machine Learning	Lab 1 — Account Safety and IAM Baseline; Lab 2 — S3 ML Data Lake and Data Formats; Lab 3 — Batch and Streaming Ingestion with Glue, Athena and Kinesis; Lab 4 — Feature Engineering, Feature Store, Labelling and Data Quality
Day 2	ML Model Development	Lab 5 — Modelling Approach, Algorithms and Managed AI Services; Lab 6 — SageMaker Training Jobs and Debugger; Lab 7 — Automatic Model Tuning, Experiments and Model Registry; Lab 8 — Model Evaluation, Metrics, Bias and Explainability
Day 3	Deployment, Monitoring, Security & Exam Review	Lab 9 — Deployment Options, Endpoints, Batch Transform and Auto Scaling; Lab 10 — MLOps Pipelines, CI/CD,

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		Orchestration and IaC; Lab 11 — Model Monitoring, Drift and Observability; Lab 12 — Security, Governance, Cost Optimisation and Exam Review

Trainer and Learner Actions

Phase	Trainer action	Learner action	Resources
Explain	Connect each AWS service to the exam domain and demonstrate the target workflow on slides.	Follow the slides, annotate the Learner Guide and note the expected outcome.	PPT, AWS documentation
Practise	Facilitate each numbered lab step and monitor cost and security guardrails.	Complete every lab step in order and retain screenshots/notes as evidence.	Learner Guide, AWS account, SageMaker AI
Validate	Check the 'Test it' result, troubleshoot and debrief design choices.	Run the validation check, capture evidence and clean up paid resources.	Lab 'Test it' and clean-up steps

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